Materials and Design 162 (2019) 418–428

Contents lists available at ScienceDirect

Materials and Design

journal homepage: www.elsevier.com/locate/matdes

Mechanical property modeling of photosensitive liquid resin in stereolithography additive manufacturing: Bridging degree of cure with tensile strength and hardness

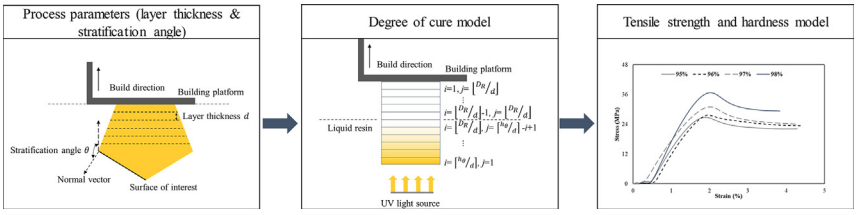
Yiran Yang, Lin Li *, Jing Zhao

Department of Mechanical and Industrial Engineering, University of Illinois at Chicago, IL 60607, USA

HIGHLIGHTS

- The tensile strength and hardness of photosensitive liquid resin are modeled based on the degree of cure.
- The established degree of cure model has an average of 94% prediction accuracy.
- The investigated resin consists of Hexane-1, 6-diol diacrylate, Isobornyl acrylate, Acrylate monomer, etc.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Received 1 October 2018
Received in revised form 29 November 2018
Accepted 3 December 2018
Available online 6 December 2018

Keywords:
Additive manufacturing
Photosensitive liquid resin
Mechanical property
Tensile strength
Hardness
Degree of cure

ABSTRACT

Owing to the unique layer-wise production method, additive manufacturing technologies have been widely adopted in rapid prototyping and tooling areas, which often require superior mechanical properties such as tensile strength and hardness. In current literature, most mechanical property studies focusing on additive manufactured materials mainly adopt experimental or simulation-based approaches, and therefore cannot be directly used to accurately estimate and predict the achieved mechanical properties. In addition, information regarding the mechanical properties of photosensitive liquid resin used in the Stereolithography additive manufacturing process is limited. Hence, in this paper, mathematical models are established to quantify the tensile strength and hardness of Stereolithography fabricated materials by estimating the solidification levels of both green parts and Ultraviolet post-cured parts. The established degree of cure model is shown to have an average prediction accuracy of around 94%. In addition, the mechanical property models have an average accuracy of 88% and 90% for tensile strength prediction, and 98% and 95% for hardness prediction of green parts and post-cured parts, respectively. It is also observed that the Ultraviolet post-curing process has the capability of significantly enhancing the studied mechanical properties.

© 2018 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

1. Introduction

Additive manufacturing (AM) fabricates objects directly from three-dimensional (3D) models by joining materials layer-by-layer. The appropriate implementation of AM technologies can help save production time and cost, shorten product development cycle, and enhance the manufacturing capability and complexity. Owing to the unique production technique, AM has been adopted in different industries such as aerospace, automobile, food, and medicine. The main applications of AM include rapid prototyping, rapid tooling, and rapid manufacturing [1]. In current literature, numerous research efforts have been conducted to enhance AM process by developing new manufacturing technologies [2,3], evaluating the environmental sustainability performance [4–7] conducting cost analysis and assessment [8,9], and improving the print accuracy and quality [10,11]. Currently, AM technologies are primarily applied in design and/or modeling stage for fit and function prototyping and tooling. In such cases, the

* Corresponding author.
E-mail address: linli@uic.edu (L. Li).

https://doi.org/10.1016/j.matdes.2018.12.009
0264-1275/© 2018 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Nomenclature

d	layer thickness (mm)
θ	stratification angle between surface normal vector and build direction (°)
θ^*	positioning angle between surface normal vector and UV post-curing light direction (°)
D_R	liquid resin depth (mm)
h_θ	part height in the build direction with certain stratification angle θ (mm)
i	index of the printed layers, $i \in [1, \lceil h_\theta/d \rceil]$
k	index of layers in UV post-curing process, $k \in [1, \lceil h_\theta/d \rceil]$
j	index of the curing times for a certain layer, $j \in [1, \lfloor D_R/d \rfloor]$
N_b	number of bottom layers
tc_i^j	curing time for the ith layer when it is being cured for the jth time (s)
tc_b	curing time of bottom layers (s)
tc_o	curing time of the rest of the layers (s)
tc^*	post-curing time (s)
t_{lift}	lift and sequence time (s)
T_i^j	temperature of the ith layer when it is being cured for the jth time (K)
T_k^*	temperature of the kth layer under UV post-curing (K)
α	average curing degree of the designed part (%)
α_i	curing degree of the ith layer (%)
α_k^*	curing degree of the kth layer in UV post-curing process (%)
α_i^j	curing degree of the ith layer when it is cured for the jth time (%)
ϕ	pre-exponential rate constant
R	gas constant (J/mol·K)
S_0	the photo-initiator concentration of the liquid resin (%)
S_i^j	photo-initiator concentration of the ith layer when it is being cured for the jth time (%)
S_k^*	photo-initiator concentration of the kth layer in UV post-curing process (%)
I_0	incident UV light intensity at the building platform surface (mW/mm^2)
I_0^*	incident UV light intensity in UV post-curing (mW/mm^2)
I_i^j	UV light intensity of the ith layer when it is being cured for the jth time (mW/mm^2)
I_k^*	UV light intensity of the kth layer in UV post-curing process (mW/mm^2)
λ	attenuation constant of the UV intensity
μ	UV absorptivity of the photo-initiator
p, q, m, n	fitting parameters in Eq. (8)
a_1, b_1, a_2, b_2	fitting parameters in Eq. (4)
$a_1^*, b_1^*, a_2^*, b_2^*$	fitting parameters in Eq. (20)

mechanical properties of prototypes or tools are crucial since the products have to withstand high pressure from the test of fitment or injection molding [12].

The mechanical properties of additive manufactured parts are not only determined by the raw material properties, but also by the AM process characteristics. To date, many research studies have been performed aiming to improve the overall mechanical performance. In current literature, the most popular approach for achieving better mechanical properties is developing new materials and/or modifying the existing AM materials with additives such as scandium-modified aluminum [13], carbon fiber reinforced thermoplastic composites [14], and nanocomposites [15]. In addition, enhanced mechanical properties can also be achieved by proper process planning activities such as choosing appropriate process parameters. In current literature, numerous research efforts have been performed to study and improve the mechanical properties of AM materials by optimizing process parameters for multiple AM processes, such as Fused Deposition Modeling (FDM) [15–30], Laser Sintering [31], and PolyJet [32].

According to current literature on mechanical properties of Stereolithography (SLA) fabricated parts [33–40], several parameters are shown to have the capability of altering the mechanical properties of printed specimens. Dulieu-Barton and Fulton studied environment variations, post-curing time periods, and batch-to-batch variations, and their influence on mechanical properties of SLA fabricated parts [33]. Chockalingam et al. conducted statistical analysis on tensile strength using Design of Experiment (DOE) methodology to investigate parameters including layer thickness, post-curing time and orientation [36]. Quintana et al. focused on the parameter build orientation, which was found out to have the ability of affecting the tensile strength [39]. Yang et al. experimentally explored the anisotropies of stiffness and strength of 3D printed polymers using the SLA process, and performed further structural designs using topology optimization aiming to enhance the mechanical behaviors [41]. During the SLA production process, the photosensitive liquid resin is solidified layer by layer. Different values of process parameters can alter the level of solidification, which further affect the mechanical behaviors of printed parts. Hence, it is critical and necessary to establish the relationships between the degree of cure and mechanical properties. Mansour et al. studied a short-term aging cycle (24 days) and concluded that the amount of curing and the mechanical property variations would reach equilibrium as aging proceeds [35]. The relationships between the degree of cure and mechanical properties are also explored by researchers including Hague et al. in [42] and Puebla et al. in [40]. These studies provided preliminary understandings on variations of mechanical properties due to different selections of AM process parameters.

The majority of the existing studies aiming to investigate the mechanical properties of additive manufactured parts by considering different process parameters mainly adopt experimental [16,17,21,25,33–36,39,40,42,43] or simulation-based [44,45] approaches. While these empirical studies can facilitate the understandings on the relations between AM process parameters and achieved mechanical properties, they fail to provide an accurate estimation and prediction on mechanical properties. To obtain a priori prediction, mathematical models are generally preferred. In current literature, a few mathematical models are developed to quantify mechanical properties. Rodríguez et al. established an analytical model for acrylonitrile butadiene styrene (ABS) materials used in the FDM process for predicting the material stiffness and strength [18]. Li et al. presented both theoretical model and experimental investigation on the mechanical properties of FDM fabricated parts, and concluded that it is possible to control the mechanical properties by adjusting deposition density and orientation [28]. To the best of the authors' knowledge, almost no mathematical models have been developed to study the mechanical properties of the SLA raw materials.

In summary, the limitations of current literature on mechanical properties of additive manufactured products are summarized as follows. The information regarding the mechanical properties of the SLA raw materials photosensitive liquid resin and its relations with process parameters are currently limited in literature, which can significantly hinder the implementations of the SLA process. In addition, the majority of existing studies on the mechanical properties of AM materials mainly use experimental and/or simulation approaches, and thus cannot be used for estimating and predicting the mechanical properties. An analytical model, which can help evaluate and estimate the mechanical properties of fabricated objects from the SLA process, is lacked in the literature. To address the above research gaps, the main goal of this paper is to establish mathematical models for bridging the degree of cure and studied mechanical properties (the ultimate tensile strength and hardness) for photosensitive liquid resin used in the SLA process. To achieve this goal, degree of curing models of both green parts (i.e., freshly

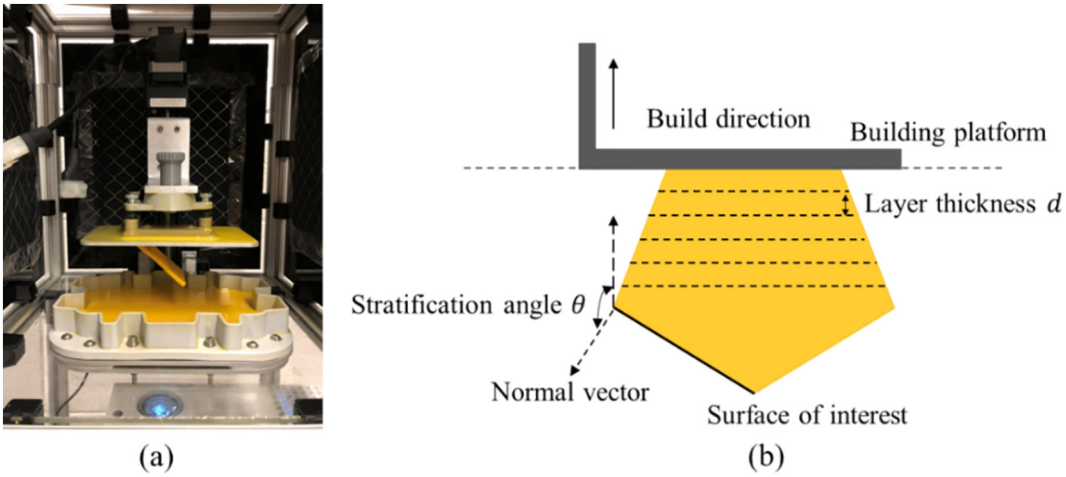


Fig. 1. Illustrations of the stereolithography process: (a) the prototype, and (b) the printed specimen.

fabricated parts) and Ultraviolet (UV) post-cured parts are proposed. The developed model is among the first a few analytical studies in the literature that can be used to estimate and predict the mechanical properties of the photosensitive liquid resin used in the SLA process. The outcomes of this paper will not only aid designers and manufacturers to obtain the desired mechanical properties by adjusting the AM process parameters in both fabrication and post-curing stages, but also facilitate the implementations of SLA fabricated parts in areas that require enhanced mechanical properties.

2. Methods

In this section, the mechanical property models are presented. More specifically, the SLA AM process is first introduced in Section 2.1, followed by the degree of cure models shown in Section 2.2 for both green parts and UV post-cured parts. Moreover, the solution algorithm for the established model is illustrated in Section 2.3. Finally, the relationships between degree of cure and achieved mechanical properties are established in Section 2.4.

2.1. Illustration of the stereolithography process

The SLA process is one of the most popular desktop AM processes, and it has obtained extensive research interest for enhancing the process performance [46–48]. As shown in Fig. 1(a), a Mask Image Projection (MIP) SLA prototype is developed and used for the purpose of this research. The process planning activities are performed in the control software, in which the 3D model is sliced into a group of 2D layer images and the values of process parameters are selected. To start the specimen fabrication, the first layer image is projected to the building platform, and then solidified by the UV light source. After the first layer is finished, the building platform moves up for the distance of layer thickness and is ready for the next layer. This "projection-solidification-movement" procedure is repeated until the entire specimen is fabricated.

2.2. Degree of cure models

During the specimen fabrication, the liquid photosensitive liquid resin is solidified by the UV light source layer-by-layer. The level of solidification (also referred to as the degree of cure) can be affected by the values of process parameters, and it can then alter the mechanical behaviors of printed specimen [35]. Therefore, to investigate the mechanical properties of printed parts and to link it with process parameters, it is critical to theoretically establish the relationships between the printing conditions and achieved degree of cure. Two process parameters are considered, i.e., layer thickness d and stratification angle θ. The layer thickness is defined as the designed height of each

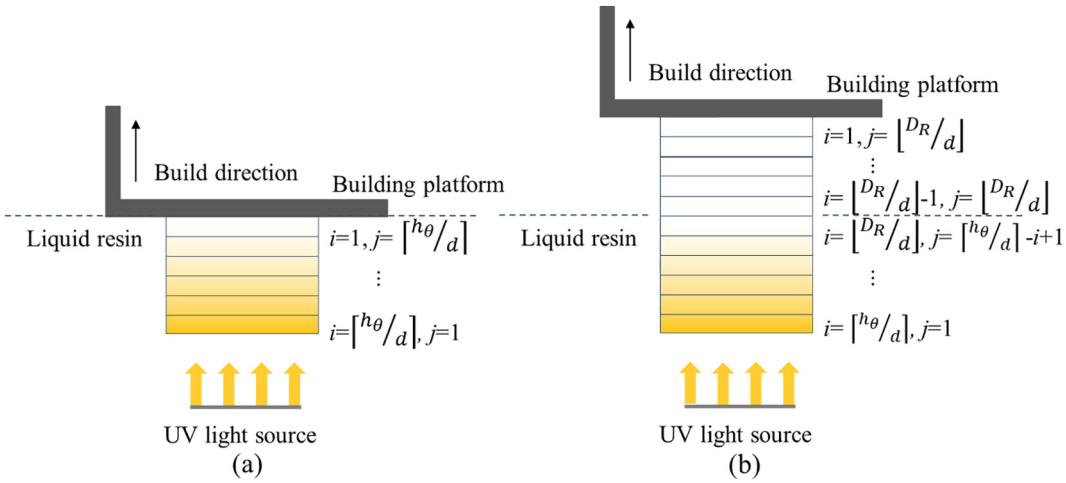


Fig. 2. Illustrations of different production scenarios: (a) $D_R \geq h_\theta$, and (b) $D_R < h_\theta$.

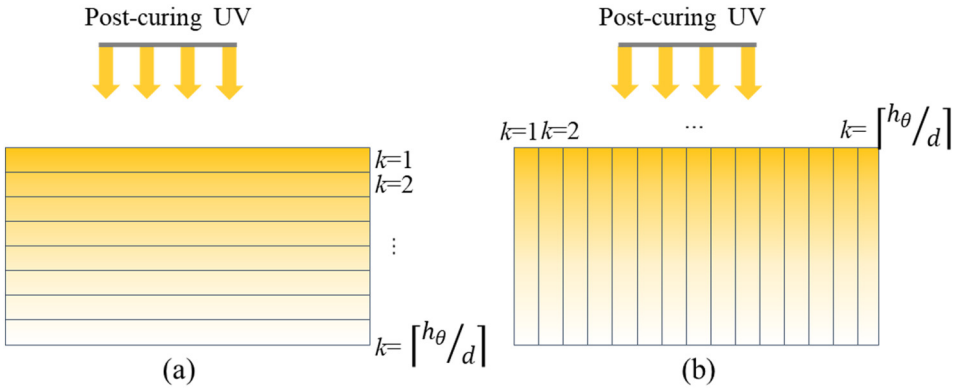


Fig. 3. Illustrations of different post-curing scenarios: (a) $\theta^* = \theta$ or $180^\circ - \theta$, and (b) $\theta^* = 90^\circ \pm \theta$.

layer. The stratification angle refers to the angle between the normal vector of surface of interest and the build direction, as illustrated in Fig. 1(b).

Two degree of cure models are established for quantifying the solidification level of green part and part after UV post-curing process, and they are shown in Sections 2.2.1 and 2.2.2, respectively.

2.2.1. Degree of cure model for green parts

During the part fabrication, two production scenarios are considered for modeling the degree of cure according to the relationship between the height of the printed part and the liquid resin depth, as illustrated in Fig. 2. In Production Scenario I, the entire part is within the liquid resin when the fabrication is finished. In other words, the liquid resin depth is larger than the part height in the building direction $D_R \geq h_\theta$. In Production Scenario II, a portion of the printed part is within the liquid resin when the fabrication is finished $D_R < h_\theta$.

Let i be the index of printed layers, $i \in [1, \lceil h_\theta/d \rceil]$. Let j be the index of curing times for a certain layer during the part fabrication process, $j \in [1, \lfloor D_R/d \rfloor]$. The consideration of index j implies that a specific layer can be solidified by the UV light for more than once, which serves as a fundamental assumption of the proposed degree of cure model. In this model, we consider a printed layer is still being slightly cured by the UV light that penetrates from the new fresh layers. This assumption also implies that the distribution of curing degree along the build direction does not remain constant. Another critical assumption of this model is that, the printed layer is assumed to be cured only when it is inside the photosensitive liquid resin. In other words, in Production Scenario I, all printed layers are continuously cured during fabrication; while in Production Scenario II, printed layers are not considered for solidification anymore after leaving the resin.

In Scenario I, $j = \lceil h_\theta/d \rceil - i + 1$. In Scenario II,

$$j = \begin{cases} \lfloor D_R/d \rfloor & i \in [1, D_R/d) \\ \lceil h_\theta/d \rceil - i + 1 & i \in [D_R/d, h_\theta/d] \end{cases} \quad (1)$$

where $\lfloor D_R/d \rfloor$ represents the floor function and $\lceil h_\theta/d \rceil$ denotes the ceiling function.

The curing time for the ith layer when it is being cured for the jth time can be formulated as:

$$tc_i^j = tc_{(i+j-1)}^1 = \begin{cases} tc_b(d) & (i+j-1) \in [1, N_b] \\ tc_o(d) & (i+j-1) \in [N_b+1, \lceil h_\theta/d \rceil] \end{cases} \quad (2)$$

where tc_b denotes the curing time for bottom layers, tc_o represents the curing time for the rest of layers of the printed part, and N_b stands for the number of bottom layers. The total production time for fabricating i number of layers can be calculated by using the following equation.

$$tp(d, \theta) = \sum_i tp_i = \sum_i \left(tc_i^{j=1} + t_{lift} \right) \quad (3)$$

In this equation, t_{lift} represents the lift and sequence time for building platform movement between fabricating layers.

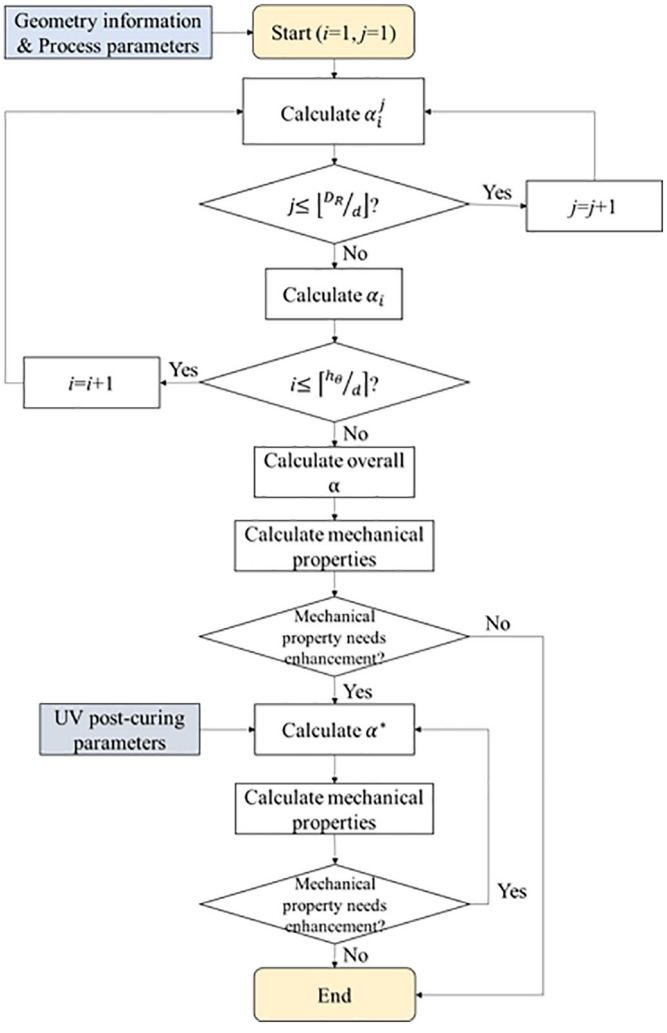


Fig. 4. The flow chart for mechanical property quantification.

Table 1
Experiment design.

Specimen No.	Layer thickness (mm)	Orientation (°)
1	0.02	0
2	0.1	
3	0.4	
4	0.02	45
5	0.1	
6	0.4	
7	0.02	90
8	0.1	
9	0.4	
10	0.08	20
11	0.14	
12	0.2	
13	0.08	40
14	0.14	
15	0.2	
16	0.08	60
17	0.14	
18	0.2	

Table 2
The experiment apparatus and specifications.

Target	Apparatus	Equipment specifications and settings
Specimen dimension	Mitutoyo 293-100 Micrometer	Resolution: 0.1 μm
Temperature profile	GSP-6 Temperature and Humidity Data Logger	Accuracy: ±0.5 °C within the range of −20 °C to 40 °C
Specimen weight	Thermogravimetric Analyzer (TGA) Q5000	Weighting resolution: 0.1 μg
Degree of cure	Differential Scanning Calorimetry (DSC) Q2000	Maximum temperature: 725 °C with the accuracy of ±0.1 °C; Heating rate of 5 °C/min from 30 °C to 300 °C
Tensile strength	Instron 3365 Tension System	Maximum load capacity: 5 kN; load speed: 0.5 mm/min
Hardness	560-10D Shore D Hardness Durometer	Range: 100HD

Table 3
Fitted parameters for temperature profiles.

Symbol	a_1	b_1	a_2	b_2
Value	300.6	2.913×10^{-14}	−3.785	-6.266×10^{-4}
Symbol	a_1^*	b_1^*	a_2^*	b_2^*
Value	303.8	2.976×10^{-7}	−7.692	-2.457×10^{-3}

During the part fabrication, the temperature levels of part and liquid resin usually vary with production time due to the use of projection. To model the temperature change, several assumptions are considered as follows. (i) Prior to the printing, the temperature levels of the SLA machine and/or components are assumed to be room temperature; (ii) During the part fabrication, the temperature of freshly printed layer is assumed to be the same as the temperature of the liquid resin at that specific time. In other words, the heat dissipated from the liquid resin to the environment is not considered in this work.

The temperature of the ith layer when it is being cured for the jth time can be formulated as:

$$T_i^j(d,\theta) = T_{(i+j-1)}^1(d,\theta) = a_1 \times \exp\left[b_1 \times \sum_{i=1}^{i=i+j-1}\left(tc_i^1 + t_{lift}\right)\right] + a_2 \times \exp\left[b_2 \times \sum_{i=1}^{i=i+j-1}\left(tc_i^1 + t_{lift}\right)\right] \quad (4)$$

In the above equation, model parameters a_1, b_1, a_2, and b_2 are determined through experimental approach based on the actual printing environment.

The degree of cure of the entire printed part is the average curing level of all the layers, and it can be formulated as follows.

$$\alpha(d,\theta) = \frac{\sum_{i=1}^{i=\lceil h_\theta/d\rceil}\alpha_i(d,\theta)}{\lceil h_\theta/d\rceil} \quad (5)$$

In this equation, $\alpha_i(d,\theta)$ denotes the curing level of the ith layer of the printed part fabricated by certain values of layer thickness and stratification angle. In Scenario I ($D_R \geq h_\theta$), $\alpha_i(d,\theta)$ can be calculated by using the following equation.

$$\alpha_i(d,\theta) = \sum_{j=1}^{j=\lceil h_\theta/d\rceil - i+1}\alpha_i^j(d,\theta) \quad (6)$$

In Scenario II ($D_R < h_\theta$), $\alpha_i(d,\theta)$ can be formulated as follows.

$$\alpha_i(d,\theta) = \begin{cases} \sum_{j=1}^{j=\lfloor D_R/d\rfloor}\alpha_i^j(d,\theta) & i\in[1, D_R/d) \\ \sum_{j=1}^{j=\lceil h_\theta/d\rceil - i+1}\alpha_i^j(d,\theta) & i\in[D_R/d, h_\theta/d] \end{cases} \quad (7)$$

The degree of cure for a specific layer i when it is solidified for the jth time can be calculated by the following equation, which is adapted from [49].

$$\alpha_i^j(d,\theta) = tc_i^j\left\{\varnothing S_i^{jq} I_i^{jp} \exp\left(\frac{-E}{RT_i^j(d,\theta)}\right)\alpha_i^j(d,\theta)^m\left[1-\alpha_i^j(d,\theta)\right]^n\right\} \quad (8)$$

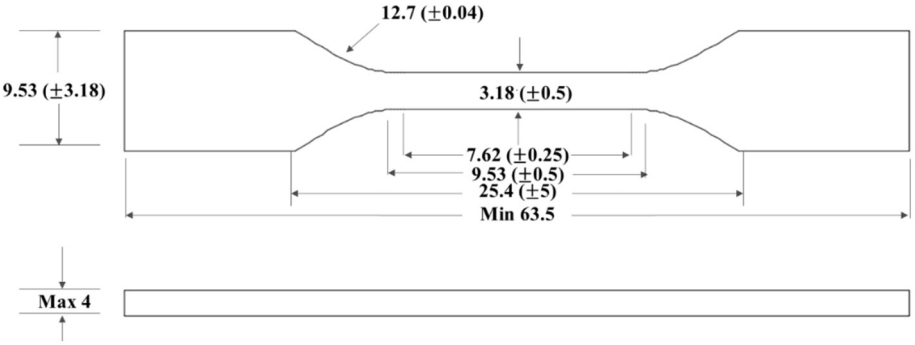


Fig. 5. Type V test specimen design specifications (unit: mm).

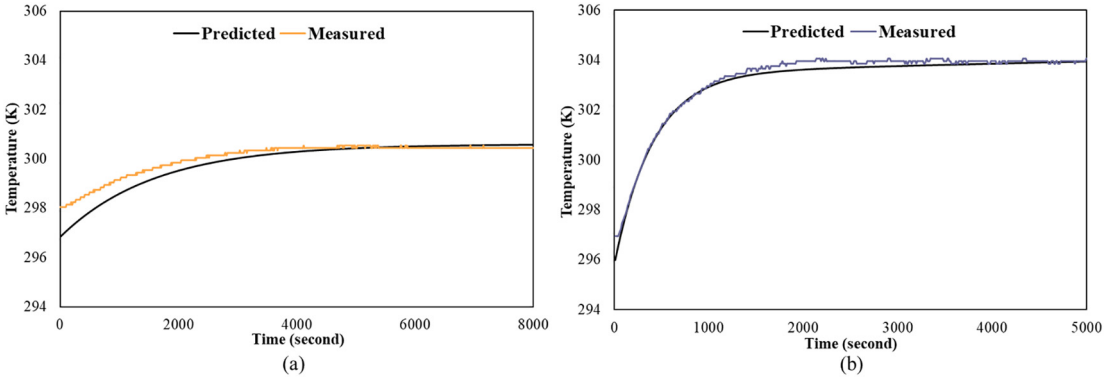


Fig. 6. The printing temperature profiles of (a) part fabrication, and (b) UV post-curing.

In this equation, S_i^j denotes the photo-initiator concentration (%) and it is assumed to decrease in reverse proportion with the curing degree. The relationship can be formulated as:

$$S_i^j = \begin{cases} S_0 & j = 1 \\ S_i^{j-1}\left[1-\alpha_i^{j-1}(d,\theta)\right] & j \geq 2 \end{cases} \tag{9}$$

S_0 represents the initial photo-initiator concentration of the liquid resin (with curing degree of zero). In addition, I_i^j denotes the light intensity (mW/mm^2) that can reach to the ith layer when it is solidified for the jth time, and it can be formulated according to Beer-Lambert law as follows.

$$I_i^j = \begin{cases} I_0 & j = 1 \\ I_0 \exp[-(j-1)\lambda d]\left[1-\sum_j \alpha_i^{j-1}(d,\theta)\right] & j \geq 2 \end{cases} \tag{10}$$

λ is the attenuation constant, and it can be calculated from $\lambda = \mu S_0$, where μ stands for the UV absorptivity of the photo-initiator [50].

To better illustrate the degree of cure model, two special cases are introduced. The first special case is when the entire specimen is fabricated by one layer; in other words, $\lceil h_\theta/d \rceil = 1$. In such case, the curing degree of the layer can be formulated as follows.

$$\begin{aligned} \alpha_1(d,\theta) &= \alpha_1^1(d,\theta) \\ &= tc_1^1\left\{\varnothing S_1^{1q} I_1^{1p} \exp\left(\frac{-E}{RT_1^1(d,\theta)}\right)\alpha_1^1(d,\theta)^m\left[1-\alpha_1^1(d,\theta)\right]^n\right\} \end{aligned} \tag{11}$$

The second special case is when the entire specimen is consisted of two layers $\lceil h_\theta/d \rceil = 2$. In this situation, the curing degree of the first layer is calculated by adding the curing degree caused by the first curing process and by the second curing process, as illustrated in the following equations.

$$\alpha_1(d,\theta) = \alpha_1^1(d,\theta) + \alpha_1^2(d,\theta) \tag{12}$$

$$\alpha_1^1(d,\theta) = tc_1^1\left\{\varnothing S_1^{1q} I_1^{1p} \exp\left(\frac{-E}{RT_1^1(d,\theta)}\right)\alpha_1^1(d,\theta)^m\left[1-\alpha_1^1(d,\theta)\right]^n\right\} \tag{13}$$

$$\alpha_1^2(d,\theta) = tc_1^2\left\{\varnothing S_1^{2q} I_1^{2p} \exp\left(\frac{-E}{RT_1^2(d,\theta)}\right)\alpha_1^2(d,\theta)^m\left[1-\alpha_1^2(d,\theta)\right]^n\right\} \tag{14}$$

The curing degree of the second layer can be formulated as follows.

$$\alpha_2(d,\theta) = \alpha_2^1(d,\theta) \tag{15}$$

$$\alpha_2^1(d,\theta) = tc_2^1\left\{\varnothing S_2^{1q} I_2^{1p} \exp\left(\frac{-E}{RT_2^1(d,\theta)}\right)\alpha_2^1(d,\theta)^m\left[1-\alpha_2^1(d,\theta)\right]^n\right\} \tag{16}$$

Hence, the overall degree of cure can be formulated as follows.

$$\alpha(d,\theta) = [\alpha_1(d,\theta) + \alpha_2(d,\theta)]/2 \tag{17}$$

2.2.2. Degree of cure model for UV post-cured parts

To enhance the surface quality and mechanical properties, additive manufactured parts usually need to be additionally processed, e.g., cleaning, UV post-curing, etc. In the proposed model, UV post-curing, as the most popular post-solidification method, is considered

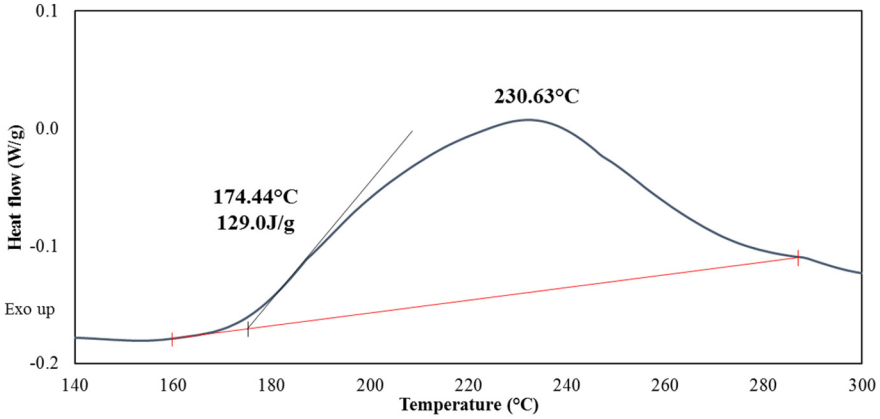


Fig. 7. The DSC heat flow plot for liquid resin ($\alpha = 0$).

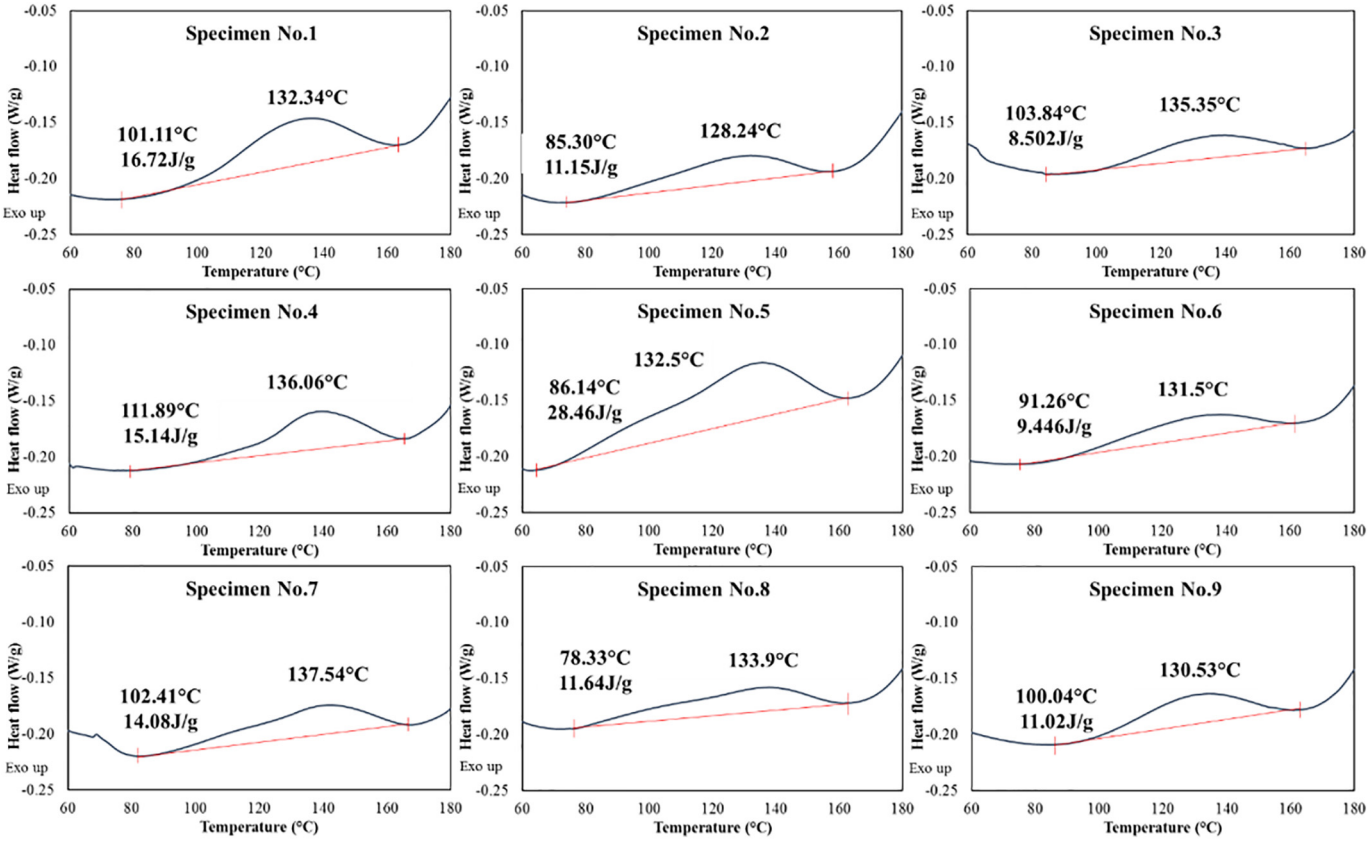


Fig. 8. The DSC heat flow plots for specimens 1–9.

in two post-curing scenarios as illustrated in Fig. 3. In Post-curing Scenario I, $\theta^* = \theta$ or $180^\circ - \theta$; in other words, the part for post-curing is positioned parallelly with the production orientation. In Post-curing Scenario II, $\theta^* = 90^\circ \pm \theta$ which indicates that the part for post-curing is positioned perpendicularly with the production orientation.

Let k be the index of post-curing layers, $k \in [1, \lceil h_\theta/d \rceil]$. The degree of cure for the kth layer in UV post-curing can be formulated as follows.

$$\alpha_k^*(d,\theta^*) = tc^*\left\{\varnothing S_k^{*q} I_k^{*p} \exp\left(\frac{-E}{RT_k^*(d,\theta^*)}\right)\alpha_k^*(d,\theta^*)^m\left[1-\alpha_k^*(d,\theta^*)\right]^n\right\} \quad (18)$$

In this equation, S_k^* denotes the photo-initiator concentration of the kth layer in the UV post-curing process, and it can be formulated as follows.

$$S_k^* = S_0[1-\alpha_k(d,\theta)] \quad (19)$$

In addition, the UV over-cuing temperature $T_k^*(d,\theta^*)$ in Eq. (18) can be formulated as follows.

$$T_k^*(d,\theta^*) = a_1^* \times \exp\left(b_1^* \times tc^*\right) + a_2^* \times \exp\left(b_2^* \times tc^*\right) \quad (20)$$

The quantification of the incident light intensity of the kth layer in the UV post-curing process I_k^* is different both two post-curing scenarios. More specifically, in Post-curing Scenario I, as shown in Fig. 3(a), $k = 1$ represents the layer that is closest to the post-curing UV light source. When $\theta^* = \theta$, $k = i$; when $\theta^* = 180 - \theta$, $k = \lceil h_\theta/d \rceil - i + 1$. I_k^* in Post-curing Scenario I can be formulated as:

$$I_k^* = I_0^* \exp[-(k-1)\lambda^* d][A(k-1)d]. \quad (21)$$

In this equation, A denotes the linear coefficient, and $\lambda^* = \mu S_k^*$. In addition, in Post-curing Scenario II, as shown in Fig. 3(b), $k = 1$ represents the layer that is in the left side of the part. In Eq. (18), $I_k^* = I_0^*$.

2.3. Degree of cure model solution algorithm

The solution algorithm for quantifying the degree of cure for part fabrication can be summarized as follows. Note that the solution algorithm for post-curing model is similar and therefore not shown in this work.

Step 1. GET values of process parameters d and θ, and geometry characteristic h_θ
Step 2. GET values of fitting parameters a_1, b_1, a_2, b_2
Step 3. CHECK if $h_\theta \le D_R$, LET $j_{max} = \lceil h_\theta/d \rceil$
ELSE, LET $j_{max} = \lfloor D_R/d \rfloor$
Step 4. GET initial values of S_0, I_0, and fitting parameters $\emptyset, E, q, p, m$, and n
FOR $i = 1: \lceil h_\theta/d \rceil$
FOR $j = 1: j_{max}$
CALCULATE T_i^j
CALCULATE α_i^j
THEN CALCULATE S_i^{j+1} and I_i^{j+1}
END LOOP
CALCULATE α_i
END LOOP
Step 5. CALCULATE α
Step 6. END

*Note that j_{max} represents the maximum value of index j.

2.4. Mechanical property models

As mentioned previously, current AM technologies are mostly used for rapid prototyping and tooling. Therefore, mechanical properties like tensile strength and hardness are crucial, since the fabricated specimens have to withstand high pressures in case of testing for fitment or

Table 4
Fitted parameters values for degree of cure models.

Symbol	∅	q	p	E	m	n
Value	0.3847	1.369	1.632	18	1.5	1.9

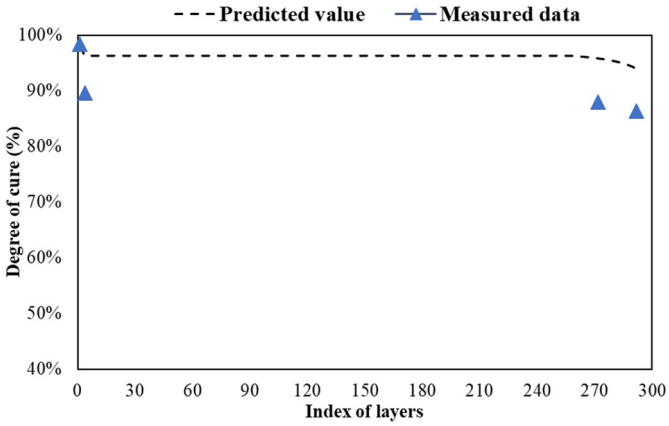


Fig. 9. The comparison between degree of cure model prediction and measurement.

injection molding. More specifically, the ultimate tensile strength (UTS), also referred to as tensile strength (TS), is used to demonstrate the material's capability of withstanding tension loads; and the hardness is used to represent the material's ability of resisting deformation induced by mechanical indentation or abrasion.

In this research, the tensile strength and hardness are studied by considering the relationships between the degree of cure and mechanical properties. The flow chart for quantifying the mechanical properties of printed specimens can be summarized in Fig. 4.

3. Results and discussions

In this section, several case studies are performed to further investigate the relationships proposed in previous section, including the degree of cure model, temperature profiles, tensile strength and hardness models.

3.1. Experiment design and apparatus

A series of experiments are designed for fitting and validating the relationships proposed in Eq. (8). More specifically, as illustrated in Table 1, specimens 1–9 are designed for obtaining the unknown values in the proposed models, and specimens 10–18 are used for model validation. Note that at least eight replicates are used for each specimen.

The specimen geometry in the experiments is designed as "Type V" according to the standard [51] proposed by ASTM International. The geometry and tolerance specifications are illustrated in Fig. 5. The specimens are fabricated using the LS600 photosensitive liquid resin by EnvisionTec [52]. The investigated resin is consisted of hexane-1,6-diol diacrylate, isobornyl acrylate, acrylate monomer, titanium dioxide, and acrylate oligomer.

To ensure that the dimensions of printed specimen conform with respect to the ASTM D638-14 standard, the dimensional integrity is examined by using an index of "precision-to-tolerance" (P/T), as formulated in the following equation.

$$P/T = k\sigma_{gauge}/(USL-LSL) \tag{22}$$

In this equation, *USL* and *LSL* denote the upper and lower specification limit, σ_{gauge} stands for the gauge deviation which can be calculated as $\sigma_{gauge} = \overline{R}/d_2$, and $k = 5.15$ for industry standard [39]. This ratio is used to investigate the variations from the measurement systems and experiment operators. The *P/T* ratio is calculated for all printed specimens. The average value is around 13.33%, which indicates that the capability of the measurement systems and inspectors is sufficient.

The experiment apparatus and corresponding specifications are listed in Table 2, including the equipment for quantifying the specimen dimension, temperature, weight, degree of cure, tensile strength, and hardness. The fabricated specimens are measured using Mitutoyo 293-100 Micrometer for their overall length, thickness, overall width, and width of the narrow section. Each dimension measurement is repeated at least eight times for better accuracy. The degree of cure of a specific specimen is quantified through Differential Scanning Calorimetry (DSC) measurement, during which the sample is heated up to 200 °C with the heating rate of 5 °C/min. The enthalpy change during this heating process is then used to calculate the solidification level of the entire specimen. The weight of the samples used in DSC test is measured using Thermogravimetric Analyzer (TGA). In addition, the temperature data (shown in Fig. 6) is obtained by recording the temperature changes during the part fabrication as well as during the UV post-curing process. More specifically, during the part fabrication, the temperature sensor is placed inside the photosensitive liquid resin. During the UV post-curing, the temperature sensor is placed on the platform in the UV post-curing chamber. In other words, the measured temperature is the farthest location from the UV light source.

The ultimate tensile strength is measured by performing tensile test using the Instron 3365 Tension System. More specifically, for each specimen, at least eight identical parts are fabricated and measured as replicates, and the averaged result is used as the ultimate tensile strength for this specific specimen. In addition, the hardness of each specimen is measured by the 560-10D Shore D Hardness Durometer with 10 readings from different locations of a specific specimen. The locations for different readings are carefully selected so that they can represent the hardness of the entire specimen. It should be noted that all the measurement data in this work has been checked for possible outliers.

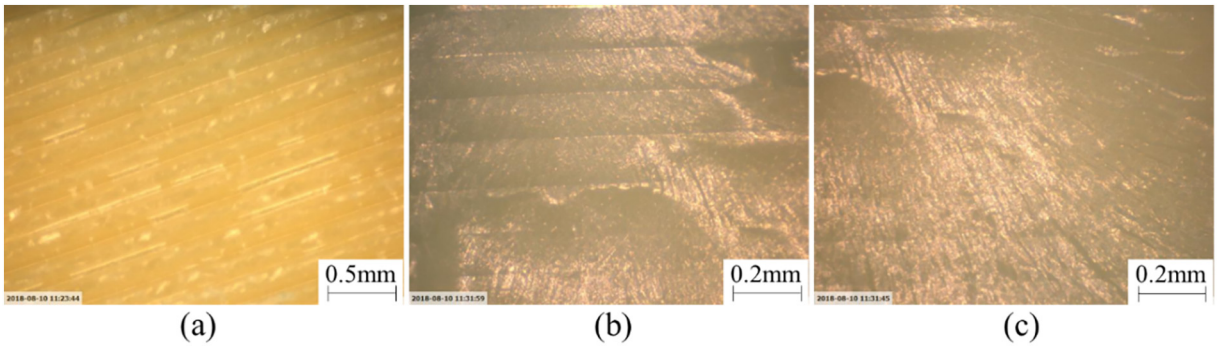


Fig. 10. The images of (a) printed surface, (b) edge of the fracture surface, and (c) center of the fracture surface.

3.2. Temperature profile analysis

In the case study, a polynomial function between the curing time and layer thickness is adopted as shown in the following equation.

$$tc_o(d) = 108.8 \times d^2 + 26.62 \times d + 1.744 \quad (23)$$

In this equation, the unit for layer thickness d is mm, and that for the curing time tc_o is second. In practice, designers and manufacturers usually use different methods to determine the appropriate curing time for each layer. For example, studies have been performed to show that the selection of curing time can affect the dimensions of printed part [53] and production temperature [54].

The temperature profiles for $T_i^j(d,\theta)$ and $T_k^*(d,\theta^*)$ are measured and fitted with R-square values of 97.62% and 99.19%, respectively. The fitted parameter values are shown in Table 3. Note that these two temperature profiles are only valid for the UV light sources studied in this work. If other UV light sources or other solidification approaches (e.g., conventional oven and microwave oven) are adopted, the temperature profiles should be remeasured.

Additional temperature measurements are obtained to compare with the predicted values, which are illustrated in Fig. 6. As shown in the plots, both predicted temperature profiles show good agreement with measured values. More specifically, model prediction accuracies of printing temperature and UV post-curing temperature are 99.92% and 99.95%, respectively.

3.3. Degree of cure model analysis

According to the DSC measurement results, the curing degree can be calculated by using the following equation.

$$\alpha(d,\theta) = \frac{\Delta H_{liquid} - \Delta H(d,\theta)}{\Delta H_{liquid}} \times 100\% \quad (24)$$

In this equation, ΔH_{liquid} denotes the enthalpy variation of the liquid resin during the heating process performed in the DSC equipment, and $\Delta H(d,\theta)$ is the enthalpy change of the measured specimen. The heat flow plot of the liquid resin is illustrated in Fig. 7. According to this plot, $\Delta H_{liquid} = 129.0$ J/g.

The DSC heat flow plots for specimens 1–9 are illustrated in Fig. 8. Note that the experimental environment and settings remain the same for all DSC measurements. As can be observed from the plots, the temperature ranges of the curing process in specimens are similar, i.e., from 60 °C to 180 °C. While the temperature ranges are similar, the enthalpy changes during the heating process significantly vary among different specimens from the lowest 8.502 J/g to the highest 28.46 J/g.

Using Eq. (24) and the DSC results, the curing degrees of specimens 1–9 are calculated and then used for fitting the model in Eq. (8). The fitted parameters are shown in Table 4.

To validate the degree of cure model, a comparison between the model prediction and DSC measurement is performed. More specifically, specimen No.14, which has a total number of 293 layers (including three bottom layers), is selected. As illustrated in Fig. 9, four different layers are chosen for DSC measurement, i.e., layer No. 1, No. 4, No. 273, and No. 293, which can represent the solidification levels from layers with different curing characteristics. As shown in the figure, a good agreement between the predicted and measured data is observed. According to the comparison, the established degree of cure model has a prediction accuracy of around 93.52%.

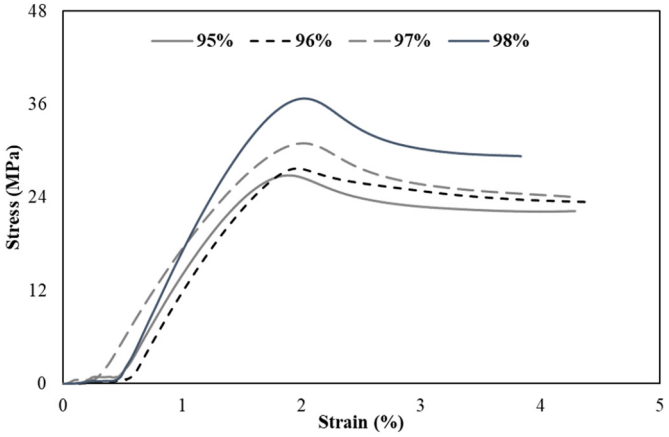


Fig. 11. The stress-strain plots for specimens with different levels of cure.

3.4. Mechanical property model analysis

Further analyses are performed in this section to investigate the mechanical properties of printed specimens. The fitted tensile strength model is as follows with R-square value of 88%.

$$UTS = 0.04747 \times \exp(2.682 \times \alpha) + 1.574 \times 10^{-14} \times \exp(36.47 \times \alpha) \quad (25)$$

To better visualize the fracture surfaces of the specimen after the tensile test, several images are obtained by using the MircoVu Precision Measurement System as shown in Fig. 10. It can be observed that around the edge of the fracture surface, the patterns of printed layers are visible; while in the center of the fracture surface, the patterns are not recognizable.

In Fig. 11, the relationships between tensile stress and tensile strain for printed specimens with different degrees of cure are demonstrated. It can be observed from the figure that a larger degree of cure can lead to a higher value of UTS. In addition, the "stress-strain" curves show different behaviors for various degrees of cure.

The performance of the established tensile strength model is shown in Table 6. According to the comparison between model predicted and measured values, the tensile strength model shows 88% accuracy in terms of estimating the tensile strength of specimens after fabrication. In addition, the model shows 90% accuracy for tensile strength estimation of specimens after UV post-curing. It can be observed from the table that green parts with smaller stratification angles tend to have lower predicted tensile strength compared to the measured values, while the opposite pattern is observed for green parts with larger stratification angles. The possible reason is that specimens with smaller stratification angles require shorter production time, which leads to underestimated temperature levels as shown in Fig. 6, and hence

Table 5
The tensile strength model accuracy.

Specimen	Tensile strength (MPa) of green parts			Tensile strength (MPa) of post-cured parts		
	Predicted	Measured	Accuracy	Predicted	Measured	Accuracy
10	26.71	36.52	73.14%	39.61	38.37	96.78%
11	26.90	33.41	80.52%	39.89	41.53	96.06%
12	26.62	30.97	85.95%	38.62	34.86	89.19%
13	27.88	25.57	90.95%	37.94	36.10	94.90%
14	27.58	26.63	96.41%	37.00	33.29	88.86%
15	27.10	24.48	89.30%	35.69	27.55	70.45%
16	28.28	24.52	84.65%	37.26	37.86	98.41%
17	27.88	27.61	99.03%	36.34	34.43	94.46%
18	27.19	25.17	91.96%	34.81	28.88	79.50%

Table 6
The hardness model accuracy.

Specimen	Hardness (Shore D) of green parts			Hardness (Shore D) of post-cured parts		
	Predicted	Measured	Accuracy	Predicted	Measured	Accuracy
10	78.36	78.21	99.81%	86.10	81.20	93.96%
11	78.47	78.60	99.83%	86.28	81.67	94.35%
12	78.31	79.24	98.82%	85.50	84.49	98.80%
13	79.04	77.93	98.57%	85.08	81.91	96.13%
14	78.87	73.28	92.38%	84.51	75.28	87.74%
15	78.58	77.16	98.16%	83.71	82.15	98.10%
16	79.27	79.51	99.70%	84.67	82.68	97.59%
17	79.04	75.45	95.25%	84.11	81.21	96.43%
18	78.64	76.21	96.82%	83.18	79.65	95.57%

cause underestimated degree of cure and tensile strength. Additionally, for the UV post-cured specimens, larger layer thickness often leads to lower tensile strength prediction accuracy. In the established degree of cure model, for a certain part, smaller layer thickness results in more accurate estimations of the UV light intensity and the photo-initiator concentration, hence leads to higher tensile strength prediction accuracy.

The fitted hardness model is as follows with R-square value of 86%.

$$Hardness = 5.148 \times 10^5 \times \exp(-9.869 \times \alpha) + 4.201 \times 10^{-9} \times \exp(23.89 \times \alpha) \quad (26)$$

To validate the proposed hardness model, comparisons between the model prediction values and experimental measurements are performed, as shown in Table 6. According to the comparison results, the hardness model has an average accuracy of 98% when estimating the hardness of green parts, and an average accuracy of 95% when estimating the hardness of UV post-cured parts. It can also be observed from the results that the hardness values of different specimens do not change significantly. The variation range is from 73 to 85 Shore D.

It can be observed from Tables 5 and 6 that the model accuracy in terms of predicting the hardness is around 8.42% higher than the model accuracy for tensile strength estimation. The possible reason is that the quantification for tensile strength has more error sources, including the quantification of the maximum load during the tensile test and the measurement of cross-section area of the specimen. In addition, the number of replications used in the hardness measurement is more than tensile test, and it can contribute to a higher prediction accuracy.

To further explore the influence of UV post-curing on mechanical properties, additional experiments are performed for comparing different time periods of UV post-curing. As illustrated in Fig. 12, the UV post-curing can alter the mechanical behaviors of the printed specimen, and it affects the tensile strength more significantly than hardness. More specifically, 2 h, 8 h, 16 h, 18 h, 20 h and 24 h of UV post-curing can enhance the tensile strength by 30.06%, 37.33%, 73.58%, 29.47%, 28.68% and − 14.03%, respectively; and improve the hardness by 2.71%, 4.67%, 8.60%, 5.76%, 5.06%, and 0.08%, respectively. It can be observed from the figure that after 16 h, both tensile strength and hardness values are decreased due to excessive exposure to UV light.

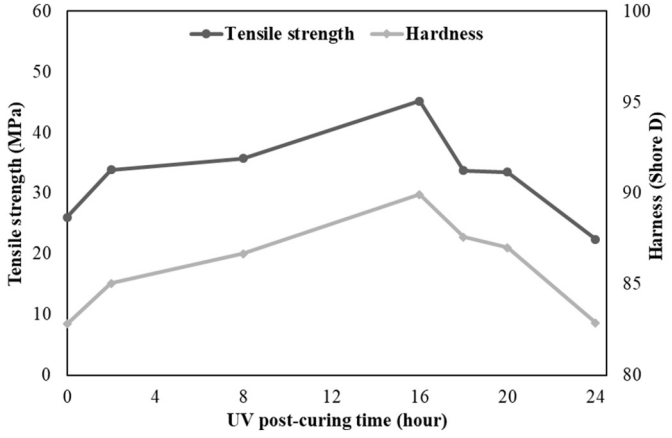


Fig. 12. Mechanical property enhancement by UV post-curing.

4. Conclusions and future work

In this work, mathematical models are established to estimate the mechanical properties of parts fabricated by the SLA process. The developed models can be used to predict the ultimate tensile strength and hardness of both green parts and UV post-cured parts. More specifically, the tensile strength model has an average accuracy of around 88% and 90% for estimating the tensile strength levels of green parts and UV post-cured parts, respectively. In addition, the hardness model shows 98% and 95% prediction accuracy for green parts and UV post-cured parts, respectively. In addition, further analysis is conducted to compare the different mechanical behaviors of UV post-cured parts due to various time periods of post-curing. The case study results show that it is possible and feasible to enhance the tensile strength and hardness using UV post-curing process.

As an extension of this work, other post-processing methods can be studied and compared with the UV light, for example, conventional oven and microwave oven. In the degree of cure model, the circular UV light source can be considered. In addition, other mechanical properties, e.g., elastic modulus, Poisson's ratio, and stiffness, can be incorporated in the mathematical models.

CRediT authorship contribution statement

Yiran Yang: Writing - original draft. **Lin Li:** Methodology. **Jing Zhao:** Data curation.

Acknowledgements

The authors truly appreciate the financial support from the U.S. National Science Foundation under Grant Number 1604825.

Data availability

The raw/processed data required to reproduce the findings in this work are uploaded.

References

[1] A. Bhargav, V. Sanjairaj, V. Rosa, L.W. Feng, J. Fuh YH, Applications of additive manufacturing in dentistry: a review, J. Biomed. Mater. Res. B Appl. Biomater. 106 (2018) 2058–2064, https://doi.org/10.1002/jbm.b.33961.
[2] L. Li, A. Haghighi, Y. Yang, A novel 6-axis hybrid additive-subtractive manufacturing process: design and case studies, J. Manuf. Process. 33 (2018) 150–160, https://doi.org/10.1016/j.jmapro.2018.05.008.
[3] L. Lu, P. Guo, Y. Pan, Magnetic-field-assisted projection stereolithography for three-dimensional printing of smart structures, J. Manuf. Sci. Eng. 139 (2017), 71008. https://doi.org/10.1115/1.4035964.
[4] S. Timothy, A. Giovanny, Z. Fu, Characterization of particle emission from fuse deposition modeling printers, ASME 2017 12th International Manufacturing Science and Engineering Conference, 2017 , (p. V002T01A040–V002T01A040).
[5] Y. Yang, L. Li, Y. Pan, Z. Sun, Energy consumption modeling of stereolithography-based additive manufacturing toward environmental sustainability, J. Ind. Ecol. (2017) https://doi.org/10.1111/jiec.12589.
[6] Y. Yang, L. Li, Total volatile organic compound emission evaluation and control for stereolithography additive manufacturing process, J. Clean. Prod. 170 (2018) 1268–1278, https://doi.org/10.1016/j.jclepro.2017.09.193.
[7] Y. Yang, A. Haghighi, L. Li, Energy consumption study for 6-axis hybrid additive-subtractive manufacturing process, Procedings of the 2018 Industrial and Systems Engineering Conference 2018, pp. 1019–1024, https://www.xcdsystem.com/iise/2018_proceedings/papers/FinalPaper_2029_0306090949.pdf.
[8] M. Baumers, L. Beltrametti, A. Gasparre, R. Hague, Informing additive manufacturing technology adoption: total cost and the impact of capacity utilisation, Int. J. Prod. Res. (2017) 1–14, https://doi.org/10.1080/00207543.2017.1334978.

[9] Y. Yang, L. Li, Cost modeling and analysis for mask image projection stereolithography additive manufacturing: simultaneous production with mixed geometries, Int. J. Prod. Econ. (2018) https://doi.org/10.1016/j.ijpe.2018.09.023.
[10] L. Li, A. Haghighi, Y. Yang, Theretical modeling and prediction of surface roughness for hybrid additive-subtractive manufacturing processes, IISE Trans. (2018) https://doi.org/10.1080/24725854.2018.1458268 (in press).
[11] B. AlMangour, J.M. Yang, Improving the surface quality and mechanical properties by shot-peening of 17-4 stainless steel fabricated by additive manufacturing, Mater. Des. 110 (2016) 914–924, https://doi.org/10.1016/j.matdes.2016.08.037.
[12] J.M. Chacón, M.A. Caminero, E. García-Plaza, P.J. Núñez, Additive manufacturing of PLA structures using fused deposition modelling: effect of process parameters on mechanical properties and their optimal selection, Mater. Des. 124 (2017) 143–157, https://doi.org/10.1016/j.matdes.2017.03.065.
[13] A.B. Spierings, K. Dawson, M. Voegtlin, F. Palm, P.J. Uggowitzer, Microstructure and mechanical properties of as-processed scandium-modified aluminium using selective laser melting, CIRP Ann. Manuf. Technol. 65 (2016) 213–216, https://doi.org/10.1016/j.cirp.2016.04.057.
[14] F. Ning, W. Cong, Y. Hu, H. Wang, Additive manufacturing of carbon fiber-reinforced plastic composites using fused deposition modeling: effects of process parameters on tensile properties, J. Compos. Mater. 51 (2017) 451–462, https://doi.org/10.1177/0021998316646169.
[15] Z. Weng, J. Wang, T. Senthil, L. Wu, Mechanical and thermal properties of ABS/montmorillonite nanocomposites for fused deposition modeling 3D printing, Mater. Des. 102 (2016) 276–283.
[16] A.K. Sood, R.K. Ohdar, S.S. Mahapatra, Parametric appraisal of mechanical property of fused deposition modelling processed parts, Mater. Des. 31 (2010) 287–295, https://doi.org/10.1016/j.matdes.2009.06.016.
[17] J.F. Rodríguez, J.P. Thomas, J.E. Renaud, Mechanical behavior of acrylonitrile butadiene styrene (ABS) fused deposition materials. Experimental investigation, Rapid Prototyp. J. 7 (2001) 148–158, https://doi.org/10.1108/13552540110395547.
[18] J.F. Rodríguez, J.P. Thomas, J.E. Renaud, Mechanical behavior of acrylonitrile butadiene styrene fused deposition materials modeling, Rapid Prototyp. J. 9 (2003) 219–230, https://doi.org/10.1108/13552540310489604.
[19] B.M. Tymrak, M. Kreiger, J.M. Pearce, Mechanical properties of components fabricated with open-source 3-D printers under realistic environmental conditions, Mater. Des. 58 (2014) 242–246, https://doi.org/10.1016/j.matdes.2014.02.038.
[20] M. Nikzad, S.H. Masood, I. Sbarski, Thermo-mechanical properties of a highly filled polymeric composites for Fused Deposition Modeling, Mater. Des. 32 (2011) 3448–3456, https://doi.org/10.1016/j.matdes.2011.01.056.
[21] M. Montero, S. Roundy, D. Odell, Material characterization of fused deposition modeling (FDM) ABS by designed experiments, Proceedings of Rapid Prototyping & Manufacturing Conference 2001, pp. 1–21 , (doi:26).
[22] M. Hossain, J. Ramos, D. Espalin, M. Perez, R. Wicker, Improving Tensile Mechanical Properties of FDM-manufactured Specimens via Modifying Build Parameters, Utwired.Engr.Utexas.Edu. 2013 380–392, http://utwired.engr.utexas.edu/lff/symposium/proceedingsarchive/pubs/manuscripts/2013/2013-30-hossain.pdf.
[23] O.S. Carneiro, A.F. Silva, R. Gomes, Fused deposition modeling with polypropylene, Mater. Des. 83 (2015) 768–776, https://doi.org/10.1016/j.matdes.2015.06.053.
[24] A.K. Sood, R.K. Ohdar, S.S. Mahapatra, Experimental investigation and empirical modelling of FDM process for compressive strength improvement, J. Adv. Res. 3 (2012) 81–90, https://doi.org/10.1016/j.jare.2011.05.001.
[25] S. Ahn, M. Montero, D. Odell, S. Roundy, P.K. Wright, Anisotropic material properties of fused deposition modeling ABS, Rapid Prototyp. J. 8 (2002) 248–257, https://doi.org/10.1108/13552540210441166.
[26] C.S. Lee, S.G. Kim, H.J. Kim, S.H. Ahn, Measurement of anisotropic compressive strength of rapid prototyping parts, J. Mater. Process. Technol. 187–188 (2007) 627–630, https://doi.org/10.1016/j.jmatprotec.2006.11.095.
[27] J.F. Rodríguez, J.P. Thomas, J.E. Renaud, Design of fused-deposition ABS components for stiffness and strength, J. Mech. Des. 125 (2003) 545, https://doi.org/10.1115/1.1582499.
[28] L. Li, Q. Sun, C. Bellehumeur, P. Gu, Composite modeling and analysis for fabrication of FDM prototypes with locally controlled properties, J. Manuf. Process. 4 (2002) 129–141, https://doi.org/10.1016/S1526-6125(02)70139-4.
[29] C. Ziemian, M. Sharma, S. Ziemian, Anisotropic mechanical properties of ABS parts fabricated by fused deposition modelling, Mech. Eng. (2012) 159–180, https://doi.org/10.5772/34233.
[30] E. Fodran, M. Koch, U. Menon, Mechanical and dimensional characteristics of fused deposition modeling build styles, Solid Freeform Fabrication Proceedings, September 1996 1996, pp. 419–442.
[31] P. Mengucci, A. Gatto, E. Bassoli, L. Denti, F. Fiori, E. Girardin, P. Bastianoni, B. Rutkowski, A. Czyrska-Filemonowicz, G. Barucca, Effects of build orientation and element partitioning on microstructure and mechanical properties of biomedical Ti-6Al-4V alloy produced by laser sintering, J. Mech. Behav. Biomed. Mater. 71 (2017) 1–9, https://doi.org/10.1016/j.jmbbm.2017.02.025.
[32] S.Y. Hong, Y.C. Kim, M. Wang, H.-I. Kim, D.-Y. Byun, J.-D. Nam, T.-W. Chou, P.M. Ajayan, L. Ci, J. Suhr, Experimental investigation of mechanical properties of UV-Curable 3D printing materials, Polymer 145 (2018) 88–94, https://doi.org/10.1016/j.polymer.2018.04.067.
[33] J.M. Dulieu-Barton, M.C. Fulton, Mechanical properties of a typical stereolithography resin, Strain 36 (2000) 81–87.
[34] G.V. Salmoria, C.H. Ahrens, M. Fredel, V. Soldi, A.T.N. Pires, Stereolithography somos 7110 resin: mechanical behavior and fractography of parts post-cured by different methods, Polym. Test. 24 (2005) 157–162, https://doi.org/10.1016/j.polymertesting.2004.09.008.
[35] S. Mansour, M. Gilbert, R. Hague, A study of the impact of short-term ageing on the mechanical properties of a stereolithography resin, Mater. Sci. Eng. A 447 (2007) 277–284, https://doi.org/10.1016/j.msea.2006.10.007.
[36] K. Chockalingam, N. Jawahar, K.N. Ramanathan, P.S. Banerjee, Optimization of stereolithography process parameters for part strength using design of experiments, Int. J. Adv. Manuf. Technol. 29 (2006) 79–88, https://doi.org/10.1007/s00170-004-2307-0.
[37] N. Saleh, N. Hopkinson, R. Hague, S. Wise, Effects of electroplating on the mechanical properties of stereolithography and laser sintered parts, Rapid Prototyp. J. 10 (2004) 305–315.
[38] J. Stampfl, S. Baudis, C. Heller, R. Liska, A. Neumeister, R. Kling, A. Ostendorf, M. Spitzbart, Photopolymers with tunable mechanical properties processed by laser-based high-resolution stereolithography, J. Micromech. Microeng. 18 (2008) https://doi.org/10.1088/0960-1317/18/12/125014.
[39] R. Quintana, J.W. Choi, K. Puebla, R. Wicker, Effects of build orientation on tensile strength for stereolithography-manufactured ASTM D-638 type I specimens, Int. J. Adv. Manuf. Technol. 46 (2010) 201–215, https://doi.org/10.1007/s00170-009-2066-z.
[40] K. Puebla, K. Arcaute, R. Quintana, R.B. Wicker, Effects of environmental conditions, aging, and build orientations on the mechanical properties of ASTM type I specimens manufactured via stereolithography, Rapid Prototyp. J. 18 (2012) 374–388, https://doi.org/10.1108/13552541211250373.
[41] K.K. Yang, J.H. Zhu, C. Wang, D.S. Jia, L.L. Song, W.H. Zhang, Experimental validation of 3D printed material behaviors and their influence on the structural topology design, Comput. Mech. 61 (2018) 581–598, https://doi.org/10.1007/s00466-018-1537-1.
[42] R. Hague, S. Mansour, N. Saleh, R. Harris, Materials analysis of stereolithography resins for use in rapid manufacturing, J. Mater. Sci. 39 (2004) 2457–2464, https://doi.org/10.1023/B:JMSC.0000020010.73768.4a.
[43] M.W. Barclift, C.B. Williams, Examining variability in the mechanical properties of parts manufactured via polyjet direct 3D printing, International Solid Freeform Fabrication Symposium 2012, pp. 876–890, https://doi.org/10.1017/CBO9781107415324.004.
[44] M. Domingo-Espin, J.M. Puigoriol-Forcada, A.A. Garcia-Granada, J. Llumà, S. Borros, G. Reyes, Mechanical property characterization and simulation of fused deposition modeling polycarbonate parts, Mater. Des. 83 (2015) 670–677, https://doi.org/10.1016/j.matdes.2015.06.074.
[45] G. Campoli, M.S. Borleffs, S. Amin Yavari, R. Wauthle, H. Weinans, A.A. Zadpoor, Mechanical properties of open-cell metallic biomaterials manufactured using additive manufacturing, Mater. Des. 49 (2013) 957–965, https://doi.org/10.1016/j.matdes.2013.01.071.
[46] H. He, Y. Pan, A. Feinerman, J. Xu, Air-diffusion-channel constrained surface based stereolithography for three-dimensional printing of objects with wide solid cross sections, J. Manuf. Sci. E. T. ASME 140 (2018) https://doi.org/10.1115/1.4039440.
[47] J.G. Zhou, D. Herscovici, C.C. Chen, Parametric process optimization to improve the accuracy of rapid prototyped stereolithography parts, Int. J. Mach. Tools Manuf. 40 (2000) 363–379, https://doi.org/10.1016/S0890-6955(99)00068-1.
[48] S.O. Onuh, K.K.B. Hon, Optimising build parameters for improved surface finish in stereolithography, Int. J. Mach. Tools Manuf. 38 (1998) 329–342, https://doi.org/10.1016/S0890-6955(97)00068-0.
[49] A. Yebi, B. Ayalew, Optimal layering time control for stepped-concurrent radiative curing process, J. Manuf. Sci. Eng. 137 (2015), 11020. https://doi.org/10.1115/1.4029023.
[50] J.M. Matias, P.J. Bartolo, A.V. Pontes, Modeling and simulation of photofabrication processes using unsaturated polyester resins, J. Appl. Polym. Sci. 114 (2009) https://doi.org/10.1002/app.30405.
[51] ASTM International, D638-14 Standard Test Method for Tensile Properties of Plastics, http://www.compass.astm.org/EDIT/html_annot.cgi?D638+14 2014.
[52] EnvisionTec, LS600, http://www.envisiontec.com/3d-printing-materials/perfactory-materials/ls600/ 2014, Accessed date: 6 February 2016.
[53] G. Mitteramskogler, R. Gmeiner, R. Felzmann, S. Gruber, C. Hofstetter, J. Stampfl, J. Ebert, W. Wachter, J. Laubersheimer, Light curing strategies for lithography-based additive manufacturing of customized ceramics, Addit. Manuf. 1 (2014) 110–118, https://doi.org/10.1016/j.addma.2014.08.003.
[54] K. Xu, Y. Chen, Photocuring temperature study for curl distortion control in projection-based stereolithography, J. Manuf. Sci. Eng. 139 (2016) https://doi.org/10.1115/1.4034305.